

# CittaEFS API Documentation

Base URL: <https://ei-api.azurewebsites.net>

## 1. Authentication

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All requests require an API Key passed via the Bearer token header.

```
Authorization: Bearer <YOUR_API_KEY>
```

### Your API Key

```
sk_live_1NZAJM5WajKYQVBo3atXDNXxM33ijmAt4Xsj7lUz
```

### ❏ Code Example (Node.js)

```
const axios = require('axios');

const response = await axios.get(
  `${ApiConfig.baseUrl}/api/einvoice/archive`,
  { headers: { 'Authorization': 'Bearer YOUR_API_KEY' } }
);
```

## 2. Webhooks Configuration

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Webhook notifications are sent as POST requests to your configured URL for the following events:

invoice.created

invoice.signed

invoice.transmitted

invoice.payment.updated

invoice.validation.failed

## Update Webhook URL

Not Configured

## Webhook Secret (HMAC-SHA256)

```
CF35DF20-9309-4506-BCC8-5D17D1DA209A
```

🔑 **Webhook Secret:** Use this secret to verify incoming webhook signatures. The secret is automatically generated when you set a webhook URL.

## 3. Receiving Webhooks

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When an event occurs, we will send a POST request to your configured webhook URL. You must verify the signature to ensure the request is from us.

### 🔑 **Webhook Payload Structure (WebhookPayload)**

```
{
  "event": "invoice.transmitted",
  "timestamp": "2026-07-28T10:30:00.000Z",
  "data": {
    "invoiceNumber": "INV2024001",
    "irn": "IRN12345678901234567890",
    "sourceNbr": "PO-001",
    "status": "TRANSMITTED",
    "qrCode": "iVBORw0KGgoAAAANSUhEUgAA...",
    "transmittedAt": "2026-07-28T10:30:00.000Z",
    "signedAt": "2026-07-28T10:25:00.000Z",
    "paymentStatus": "PENDING",
    "paymentReference": null,
    "totalAmount": 1875000.00,
    "currency": "NGN",
    "errorMessage": null,
    "eventType": "invoice.transmitted",
    "timestamp": "2026-07-28T10:30:00.000Z",
    "metadata": {
      "CustomField1": "Value1",
      "CustomField2": "Value2"
    }
  },
  "webhookMetadata": {
    "companyId": "550e8400-e29b-41d4-a716-446655440000",
    "webhookId": "123e4567-e89b-12d3-a456-426614174000",
    "attempt": 1
  }
}
```

## 🔒 Signature Verification

Each webhook request includes an `X-Webhook-Signature` header. Use your webhook secret to verify the signature using HMAC-SHA256:

```
X-Webhook-Signature: sha256=abcdef1234567890...
```

## 🔒 Code Examples (Node.js)

```

const crypto = require('crypto');
const express = require('express');
const app = express();

const WEBHOOK_SECRET = process.env.WEBHOOK_SECRET;

app.use(express.raw({ type: 'application/json' }));

app.post('/webhook', (req, res) => {
  const signature = req.headers['x-webhook-signature'];
  if (!signature) {
    return res.status(400).send('Missing signature');
  }

  const payload = req.body.toString();

  const hmac = crypto.createHmac('sha256', WEBHOOK_SECRET);
  const computedHash = hmac.update(payload).digest('base64');
  const expectedSignature = `sha256=${computedHash}`;

  if (signature !== expectedSignature) {
    return res.status(401).send('Invalid signature');
  }

  const data = JSON.parse(payload);
  const event = data.event;

  switch (event) {
    case 'invoice.transmitted':
      console.log(`Invoice ${data.data.invoiceNumber} transmitted`);
      break;
    case 'invoice.payment.updated':
      console.log(`Payment updated for ${data.data.invoiceNumber}`);
      break;
  }

  res.json({ status: 'ok' });
});

```

## 4. Submit E-Invoices

---

**POST** /api/integration/gen/invoices  
 Submit invoice line items directly.

☒ **Request Payload (IntegrationInvoiceLineDto)**

```
[
  {
    // ===== REQUIRED FIELDS =====
    "invoiceNumber": "INV2024001",
    "issueDate": "2026-07-28",
    "customerCode": "CUST001",
    "itemName": "Laptop Pro",
    "itemDescription": "15-inch Laptop with 16GB RAM, 512GB SSD",
    "quantity": 1,
    "unitPrice": 1500000.00,
    "taxAmount": 225000.00,
    "taxableAmount": 1500000.00,
    "hsOrServiceCode": "2620",

    // ===== OPTIONAL FIELDS =====
    "lineNum": "1",
    "unitCode": "EA",
    "taxCategoryId": "STANDARD_VAT",
    "currencyCode": "NGN",
    "invoiceTypeCode": "381",
    "headerDiscount": 0.00,
    "headerCharges": 0.00,
    "lineDiscount": 0.00,
    "useStateTax": false,
    "documentNumber": "PO-001",
    "billingReferenceIrns": ["IRN12345678901234567890"],

    // ===== CUSTOM FIELDS =====
    "customFields": {
      "Department": "IT",
      "CostCenter": "CC-001",
      "ProjectCode": "PRJ-2024-001"
    },

    // ===== METADATA =====
    "metadata": {
      "SourceSystem": "ERP",
      "UserID": "user123",
      "BranchCode": "LAGOS-001",
      "SalesPerson": "John Doe"
    }
  }
]
```

## ❏ Response (BulkEInvoiceResultDto)

```
{
  "totalInvoices": 1,
  "successCount": 1,
  "failedCount": 0,
  "errors": []
}
```

### ❏ Code Example (Node.js)

```
const payload = [{
  invoiceNumber: "INV2024001",
  issueDate: "2026-07-28",
  customerCode: "CUST001",
  itemName: "Laptop Pro",
  itemDescription: "Laptop Pro",
  quantity: 1,
  unitPrice: 1500000,
  taxAmount: 225000,
  taxableAmount: 1500000,
  hsOrServiceCode: "84713000"
}];

const response = await axios.post(
  `${ApiConfig.baseUrl}/api/integration/gen/invoices`,
  payload,
  { headers: { 'Authorization': 'Bearer YOUR_API_KEY' } }
);
```

## 5. Get Archive Records

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**GET** /api/einvoice/archive  
Fetch successfully transmitted invoices.

### ❏ Query Parameters:

| Field         | Type   | Presence | Description             |
|---------------|--------|----------|-------------------------|
| fromDate      | string | Optional | Start Date (YYYY-MM-DD) |
| toDate        | string | Optional | End Date (YYYY-MM-DD)   |
| paymentStatus | string | Optional | PAID, UNPAID, PARTIAL   |

### ❏ Response (ArchiveResponseDto)

```
[
  {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "originalInvoiceId": "550e8400-e29b-41d4-a716-446655440001",
    "seqNo": 1,
    "invoiceNumber": "INV2024001",
    "irn": "IRN12345678901234567890",
    "qrCode": "iVBORw0KGgoAAAANSUhEUgAA...",
    "transmittedAt": "2026-07-28T10:30:00.000Z",
    "paymentStatus": "PAID",
    "paymentReference": "BANKREF123456",
    "paymentUpdatedAt": "2026-07-29T14:30:00.000Z",
    "customFields": {
      "Department": "IT",
      "CostCenter": "CC-001"
    },
    "metadata": {
      "SourceSystem": "ERP",
      "UserID": "user123"
    }
  }
]
```

## 🔗 Code Example (Node.js)

```
const response = await axios.get(
  `${ApiConfig.baseUrl}/api/einvoice/archive`,
  {
    params: {
      fromDate: '2026-07-01',
      toDate: '2026-07-28',
      paymentStatus: 'PAID'
    },
    headers: { 'Authorization': 'Bearer YOUR_API_KEY' }
  }
);
```

## 6. Error Monitoring

---

**GET** /api/einvoice/errors/validation  
Get validation failures.

**GET** /api/einvoice/errors/transmit  
Get transmission failures.

**GET**`/api/einvoice/errors/sign`

Get signature failures.

## ❌ Validation Error Response (ValidationErrorResponseDto)

```
[
  {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "invoiceNumber": "INV2024001",
    "rowNumber": 1,
    "fieldName": "TaxableAmount",
    "expectedValue": "1500.00",
    "actualValue": "150.00",
    "errorMessage": "TaxableAmount must equal (UnitPrice × Quantity -
LineDiscount)",
    "createdAt": "2026-07-28T10:30:00.000Z",
    "originalData":
    "{\\"invoiceNumber\\":\\"INV2024001\\",\\"rowNumber\\":1,\\"field\\":\\"TaxableAmount\\"}"
  },
  {
    "metadata": {
      "SourceSystem": "ERP"
    }
  }
]
```

## ❌ Sign Error Response (SignErrorResponseDto)

```
[
  {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "seqNo": 1,
    "invoiceNumber": "INV2024001",
    "errorMessage": "Certificate validation failed",
    "createdAt": "2026-07-28T10:30:00.000Z",
    "originalInvoiceJson":
    "{\\"business_id\\":\\"123\\",\\"invoice_number\\":\\"INV2024001\\"}"
  }
]
```

## ❌ Transmit Error Response (TransmitErrorResponseDto)

```
[
  {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "originalInvoiceId": "550e8400-e29b-41d4-a716-446655440001",
    "seqNo": 1,
    "invoiceNumber": "INV2024001",
    "irn": "IRN12345678901234567890",
    "errorMessage": "Transmission timeout",
    "createdAt": "2026-07-28T10:30:00.000Z"
  }
]
```

## ❏ Code Example (Node.js)

```
const response = await axios.get(
  `${ApiConfig.baseUrl}/api/einvoice/errors/validation`,
  {
    params: { fromDate: '2026-07-01', toDate: '2026-07-28' },
    headers: { 'Authorization': 'Bearer YOUR_API_KEY' }
  }
);
```

## 7. Payment Management

---

**PATCH** /api/einvoice/update/{irn}

Update single invoice payment.

**PATCH** /api/einvoice/bulk/update

Bulk update payments.

## ❏ Single Update Payload (UpdatePaymentStatusDto)

```
{
  "payment_status": "PAID",    // PAID | REJECTED | PARTIAL
  "reference": "BANKREF123456"
}
```

## ❏ Single Update Response (PaymentStatusResponseDto)

```
{
  "success": true,
  "message": "Payment status updated successfully",
  "invoiceNumber": "INV2024001",
  "irn": "IRN12345678901234567890",
  "paymentStatus": "PAID",
  "reference": "BANKREF123456",
  "updatedAt": "2026-07-29T14:30:00.000Z"
}
```

## ❏ Bulk Update Payload (BulkUpdatePaymentStatusDto)

```
{
  "items": [
    {
      "irn": "IRN12345678901234567890",
      "payment_status": "PAID",
      "reference": "BANKREF123456"
    },
    {
      "irn": "IRN09876543210987654321",
      "payment_status": "PARTIAL",
      "reference": "BANKREF789012"
    },
    {
      "irn": "IRN11223344556677889900",
      "payment_status": "REJECTED",
      "reference": "BANKREF345678"
    }
  ]
}
```

## ❏ Bulk Update Response (BulkPaymentStatusResponseDto)

```
{
  "success": true,
  "message": "Successfully updated 3 invoices",
  "total": 3,
  "successCount": 3,
  "failedCount": 0,
  "results": [
    {
      "irn": "IRN12345678901234567890",
      "invoiceNumber": "INV2024001",
      "paymentStatus": "PAID",
      "reference": "BANKREF123456",
      "success": true,
      "message": "Payment status updated successfully"
    },
    {
      "irn": "IRN09876543210987654321",
      "invoiceNumber": "INV2024002",
      "paymentStatus": "PARTIAL",
      "reference": "BANKREF789012",
      "success": true,
      "message": "Payment status updated successfully"
    },
    {
      "irn": "IRN11223344556677889900",
      "invoiceNumber": "INV2024003",
      "paymentStatus": "REJECTED",
      "reference": "BANKREF345678",
      "success": true,
      "message": "Payment status updated successfully"
    }
  ],
  "errors": []
}
```

## ❏ Code Example (Node.js)

```
const payload = {
  items: [{
    irn: "IRN1234567890",
    payment_status: "PAID",
    reference: "BANKREF123"
  }]
};

const response = await axios.patch(
  `${ApiConfig.baseUrl}/api/einvoice/bulk/update`,
  payload,
  { headers: { 'Authorization': 'Bearer YOUR_API_KEY' } }
);
```